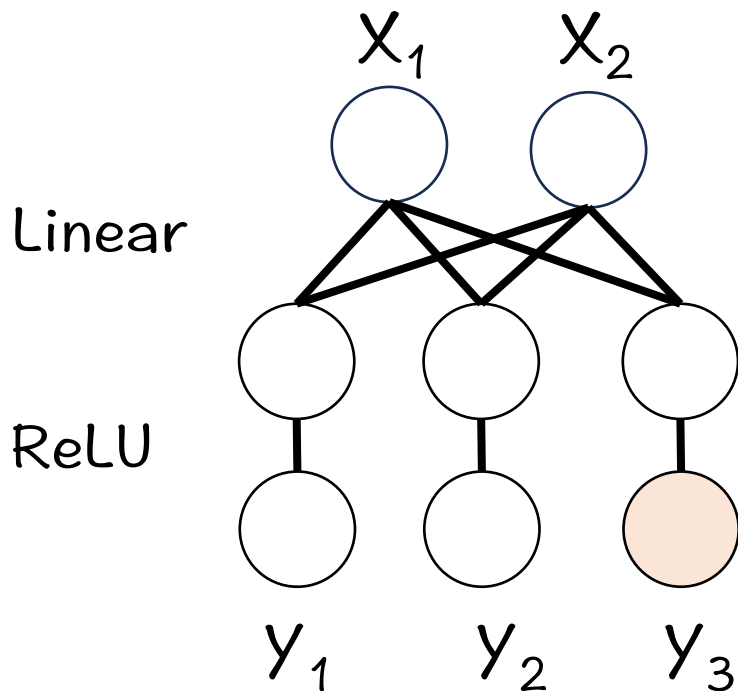


# Activation (ReLU, Sigmoid, Tanh)

## Exercise 1

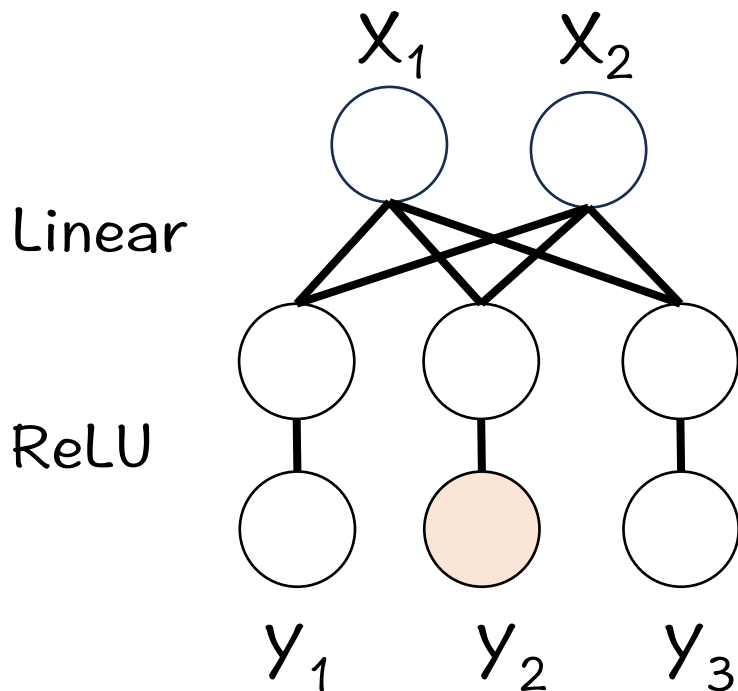
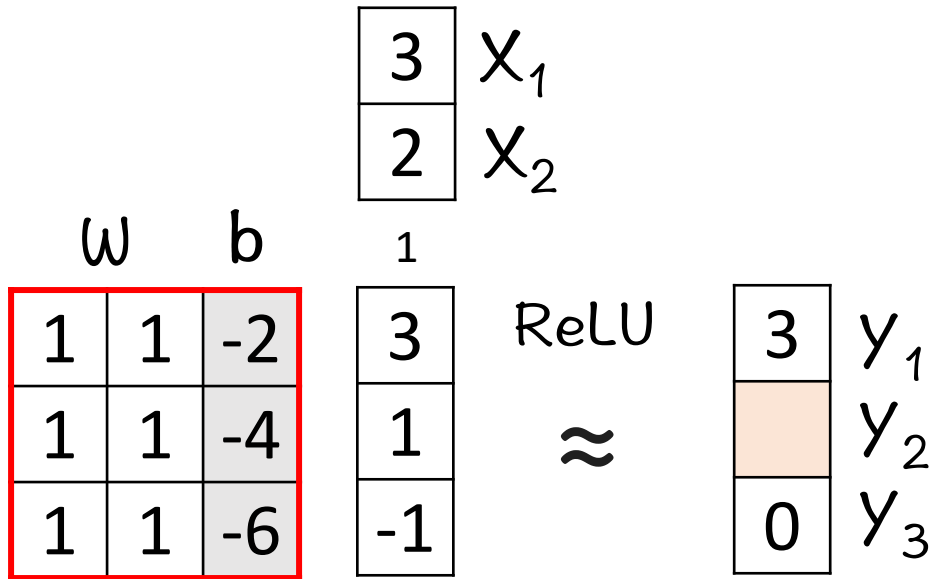
|     |     |   |    |           |   |       |       |
|-----|-----|---|----|-----------|---|-------|-------|
|     |     |   | 3  | $x_1$     |   |       |       |
|     |     |   | 2  | $x_2$     |   |       |       |
| $w$ | $b$ |   | 1  |           |   |       |       |
| 1   | -1  | 0 | 1  | ReLU      | 1 | $y_1$ |       |
| 1   | -2  | 0 | -1 | $\approx$ | 0 | $y_2$ |       |
| 1   | -3  | 0 | -3 |           |   |       | $y_3$ |



0

# Activation (ReLU, Sigmoid, Tanh)

## Exercise 2



# Activation (ReLU, Sigmoid, Tanh)

## Exercise 3

|    |           |  |
|----|-----------|--|
| 5  | ReLU      |  |
| 4  |           |  |
| 3  | $\approx$ |  |
| 2  |           |  |
| 1  |           |  |
| 0  |           |  |
| -1 |           |  |
| -2 |           |  |
| -3 |           |  |
| -4 |           |  |
| -5 |           |  |

# Activation (ReLU, Sigmoid, Tanh)

## Exercise 4

|    |           |   |
|----|-----------|---|
| 2  | ReLU      | 2 |
| 0  | $\approx$ | 0 |
| -2 |           |   |

0

# Activation (ReLU, Sigmoid, Tanh)

## Exercise 5

|    |                                       |   |
|----|---------------------------------------|---|
| 2  | Sigmoid<br>$\sigma\{3\}$<br>$\approx$ | 1 |
| 0  |                                       |   |
| -2 |                                       | 0 |

3 Level Quantization  
for hand calculation 🖋️

| Integer   | Sigmoid |
|-----------|---------|
| $\geq 2$  | 1       |
| -1, 0, 1  | 0.5     |
| $\leq -2$ | 0       |

0.5

# Activation (ReLU, Sigmoid, Tanh)

## Exercise 6

|    |                                       |     |
|----|---------------------------------------|-----|
| -3 | Sigmoid<br>$\sigma\{3\}$<br>$\approx$ | 0   |
| 1  |                                       |     |
| -1 |                                       | 0.5 |

3 Level Quantization  
for hand calculation 🖋️

| Integer   | Sigmoid |
|-----------|---------|
| $\geq 2$  | 1       |
| -1, 0, 1  | 0.5     |
| $\leq -2$ | 0       |

0.5

# Activation (ReLU, Sigmoid, Tanh)

## Exercise 7

|    |               |   |
|----|---------------|---|
| -9 | Sigmoid       | 0 |
| -5 | $\sigma\{3\}$ |   |
| -4 | $\approx$     | 0 |

3 Level Quantization  
for hand calculation 🖋️

| Integer   | Sigmoid |
|-----------|---------|
| $\geq 2$  | 1       |
| -1, 0, 1  | 0.5     |
| $\leq -2$ | 0       |

0

# Activation (ReLU, Sigmoid, Tanh)

## Exercise 8

|    |                                       |   |
|----|---------------------------------------|---|
| 11 | Sigmoid<br>$\sigma\{3\}$<br>$\approx$ | 1 |
| 7  |                                       |   |
| 4  |                                       | 1 |

3 Level Quantization  
for hand calculation 🖋️

| Integer   | Sigmoid |
|-----------|---------|
| $\geq 2$  | 1       |
| -1, 0, 1  | 0.5     |
| $\leq -2$ | 0       |



# Activation (ReLU, Sigmoid, Tanh)

## Exercise 9

|     |     |   |                                                       |                                       |     |       |  |  |  |
|-----|-----|---|-------------------------------------------------------|---------------------------------------|-----|-------|--|--|--|
|     |     |   | <table><tr><td>3</td></tr><tr><td>2</td></tr></table> | 3                                     | 2   | $x_1$ |  |  |  |
| 3   |     |   |                                                       |                                       |     |       |  |  |  |
| 2   |     |   |                                                       |                                       |     |       |  |  |  |
|     |     |   |                                                       | $x_2$                                 |     |       |  |  |  |
| $w$ | $b$ |   | 1                                                     | Sigmoid<br>$\sigma\{3\}$<br>$\approx$ |     |       |  |  |  |
| 1   | -1  | 0 | 1                                                     |                                       | 0.5 | $y_1$ |  |  |  |
| 1   | -2  | 0 | -1                                                    |                                       |     | $y_2$ |  |  |  |
| 1   | -3  | 0 | -3                                                    |                                       | 0   | $y_3$ |  |  |  |

3 Level Quantization  
for hand calculation 🖋️

| Integer   | Sigmoid |
|-----------|---------|
| $\geq 2$  | 1       |
| -1, 0, 1  | 0.5     |
| $\leq -2$ | 0       |

0.5

# Activation (ReLU, Sigmoid, Tanh)

## Exercise 10

|     |     |   |   |                                       |     |       |  |
|-----|-----|---|---|---------------------------------------|-----|-------|--|
|     |     |   | 3 | $x_1$                                 |     |       |  |
|     |     |   | 2 | $x_2$                                 |     |       |  |
| $w$ | $b$ |   | 1 | Sigmoid<br>$\sigma\{3\}$<br>$\approx$ |     |       |  |
| 1   | -1  | 3 | 4 |                                       |     | $y_1$ |  |
| 1   | -2  | 3 | 2 |                                       | 1   | $y_2$ |  |
| 1   | -3  | 3 | 0 |                                       | 0.5 | $y_3$ |  |

3 Level Quantization  
for hand calculation 🖋️

| Integer   | Sigmoid |
|-----------|---------|
| $\geq 2$  | 1       |
| -1, 0, 1  | 0.5     |
| $\leq -2$ | 0       |

1

# Activation (ReLU, Sigmoid, Tanh)

## Exercise 11

|     |     |    |    |               |         |       |  |
|-----|-----|----|----|---------------|---------|-------|--|
|     |     |    | 3  | $x_1$         |         |       |  |
|     |     |    | 2  | $x_2$         |         |       |  |
| $w$ | $b$ |    | 1  |               | Sigmoid |       |  |
| 1   | -1  | -2 | -1 | $\sigma\{3\}$ | 0.5     | $y_1$ |  |
| 1   | -2  | -2 | -3 | $\approx$     |         | $y_2$ |  |
| 1   | -3  | -2 | -5 |               | 0       | $y_3$ |  |

3 Level Quantization  
for hand calculation 🖋️

| Integer   | Sigmoid |
|-----------|---------|
| $\geq 2$  | 1       |
| -1, 0, 1  | 0.5     |
| $\leq -2$ | 0       |

0

# Activation (ReLU, Sigmoid, Tanh)

## Exercise 12

$$\begin{array}{|c|} \hline 2 \\ \hline 0 \\ \hline -2 \\ \hline \end{array} \quad \text{Tanh}\{3\} \quad \begin{array}{|c|} \hline 1 \\ \hline \text{ } \\ \hline -1 \\ \hline \end{array} \approx$$

3 Level Quantization  
for hand calculation 🖋️

| Integer   | Sigmoid | Tanh |
|-----------|---------|------|
| $\geq 2$  | 1       | 1    |
| -1, 0, 1  | 0.5     | 0    |
| $\leq -2$ | 0       | -1   |

0

# Activation (ReLU, Sigmoid, Tanh)

## Exercise 13

|    |                           |    |
|----|---------------------------|----|
| -9 | $\tanh\{3\}$<br>$\approx$ | -1 |
| -5 |                           |    |
| -4 |                           | -1 |

3 Level Quantization  
for hand calculation 🖋️

| Integer   | Sigmoid | Tanh |
|-----------|---------|------|
| $\geq 2$  | 1       | 1    |
| -1, 0, 1  | 0.5     | 0    |
| $\leq -2$ | 0       | -1   |

-1

# Activation (ReLU, Sigmoid, Tanh)

## Exercise 14

|    |
|----|
| 11 |
| 7  |
| 4  |

 $\text{Tanh}\{3\}$   
 $\approx$ 

|   |
|---|
| 1 |
|   |
| 1 |

3 Level Quantization  
for hand calculation 🖋️

| Integer   | Sigmoid | Tanh |
|-----------|---------|------|
| $\geq 2$  | 1       | 1    |
| -1, 0, 1  | 0.5     | 0    |
| $\leq -2$ | 0       | -1   |

1

# Activation (ReLU, Sigmoid, Tanh)

## Exercise 15

|                                                                                                                                         |     |    |                                                       |   |    |                |   |    |   |  |                                                                            |   |    |    |                                 |                                                                          |   |  |    |                         |
|-----------------------------------------------------------------------------------------------------------------------------------------|-----|----|-------------------------------------------------------|---|----|----------------|---|----|---|--|----------------------------------------------------------------------------|---|----|----|---------------------------------|--------------------------------------------------------------------------|---|--|----|-------------------------|
|                                                                                                                                         |     |    | <table><tr><td>3</td></tr><tr><td>2</td></tr></table> | 3 | 2  | $x_1$<br>$x_2$ |   |    |   |  |                                                                            |   |    |    |                                 |                                                                          |   |  |    |                         |
| 3                                                                                                                                       |     |    |                                                       |   |    |                |   |    |   |  |                                                                            |   |    |    |                                 |                                                                          |   |  |    |                         |
| 2                                                                                                                                       |     |    |                                                       |   |    |                |   |    |   |  |                                                                            |   |    |    |                                 |                                                                          |   |  |    |                         |
| $w$                                                                                                                                     | $b$ | 1  |                                                       |   |    |                |   |    |   |  |                                                                            |   |    |    |                                 |                                                                          |   |  |    |                         |
| <table><tr><td>1</td><td>-1</td><td>0</td></tr><tr><td>1</td><td>-2</td><td>0</td></tr><tr><td>1</td><td>-3</td><td>0</td></tr></table> | 1   | -1 | 0                                                     | 1 | -2 | 0              | 1 | -3 | 0 |  | <table><tr><td>1</td></tr><tr><td>-1</td></tr><tr><td>-3</td></tr></table> | 1 | -1 | -3 | $\text{Tanh}\{3\}$<br>$\approx$ | <table><tr><td>0</td></tr><tr><td></td></tr><tr><td>-1</td></tr></table> | 0 |  | -1 | $y_1$<br>$y_2$<br>$y_3$ |
| 1                                                                                                                                       | -1  | 0  |                                                       |   |    |                |   |    |   |  |                                                                            |   |    |    |                                 |                                                                          |   |  |    |                         |
| 1                                                                                                                                       | -2  | 0  |                                                       |   |    |                |   |    |   |  |                                                                            |   |    |    |                                 |                                                                          |   |  |    |                         |
| 1                                                                                                                                       | -3  | 0  |                                                       |   |    |                |   |    |   |  |                                                                            |   |    |    |                                 |                                                                          |   |  |    |                         |
| 1                                                                                                                                       |     |    |                                                       |   |    |                |   |    |   |  |                                                                            |   |    |    |                                 |                                                                          |   |  |    |                         |
| -1                                                                                                                                      |     |    |                                                       |   |    |                |   |    |   |  |                                                                            |   |    |    |                                 |                                                                          |   |  |    |                         |
| -3                                                                                                                                      |     |    |                                                       |   |    |                |   |    |   |  |                                                                            |   |    |    |                                 |                                                                          |   |  |    |                         |
| 0                                                                                                                                       |     |    |                                                       |   |    |                |   |    |   |  |                                                                            |   |    |    |                                 |                                                                          |   |  |    |                         |
|                                                                                                                                         |     |    |                                                       |   |    |                |   |    |   |  |                                                                            |   |    |    |                                 |                                                                          |   |  |    |                         |
| -1                                                                                                                                      |     |    |                                                       |   |    |                |   |    |   |  |                                                                            |   |    |    |                                 |                                                                          |   |  |    |                         |

3 Level Quantization  
for hand calculation 🖋️

| Integer   | Sigmoid | Tanh |
|-----------|---------|------|
| $\geq 2$  | 1       | 1    |
| -1, 0, 1  | 0.5     | 0    |
| $\leq -2$ | 0       | -1   |

0

# Activation (ReLU, Sigmoid, Tanh)

## Exercise 16

$$\begin{array}{cc} \begin{array}{cc} w & b \\ \begin{array}{|c|c|c|} \hline 1 & -1 & 3 \\ \hline 1 & -2 & 3 \\ \hline 1 & -3 & 3 \\ \hline \end{array} & \begin{array}{|c|} \hline 1 \\ \hline \end{array} \\ \end{array} & \begin{array}{|c|} \hline 3 \\ \hline 2 \\ \hline \end{array} \begin{array}{l} x_1 \\ x_2 \end{array} \end{array} \quad \begin{array}{|c|} \hline 4 \\ \hline 2 \\ \hline 0 \\ \hline \end{array} \quad \begin{array}{l} \text{Tanh}\{3\} \\ \approx \end{array} \quad \begin{array}{|c|} \hline \text{ } \\ \hline 1 \\ \hline 0 \\ \hline \end{array} \begin{array}{l} y_1 \\ y_2 \\ y_3 \end{array}$$

3 Level Quantization  
for hand calculation 🖋️

| Integer   | Sigmoid | Tanh |
|-----------|---------|------|
| $\geq 2$  | 1       | 1    |
| -1, 0, 1  | 0.5     | 0    |
| $\leq -2$ | 0       | -1   |

1



# Activation (ReLU, Sigmoid, Tanh)

## Exercise 17

|                                                                                                                                            |     |    |                                                       |   |    |                |   |    |    |  |                                                                             |    |    |    |              |                                                                          |   |  |    |                         |
|--------------------------------------------------------------------------------------------------------------------------------------------|-----|----|-------------------------------------------------------|---|----|----------------|---|----|----|--|-----------------------------------------------------------------------------|----|----|----|--------------|--------------------------------------------------------------------------|---|--|----|-------------------------|
|                                                                                                                                            |     |    | <table><tr><td>3</td></tr><tr><td>2</td></tr></table> | 3 | 2  | $x_1$<br>$x_2$ |   |    |    |  |                                                                             |    |    |    |              |                                                                          |   |  |    |                         |
| 3                                                                                                                                          |     |    |                                                       |   |    |                |   |    |    |  |                                                                             |    |    |    |              |                                                                          |   |  |    |                         |
| 2                                                                                                                                          |     |    |                                                       |   |    |                |   |    |    |  |                                                                             |    |    |    |              |                                                                          |   |  |    |                         |
| $w$                                                                                                                                        | $b$ | 1  |                                                       |   |    |                |   |    |    |  |                                                                             |    |    |    |              |                                                                          |   |  |    |                         |
| <table><tr><td>1</td><td>-1</td><td>-2</td></tr><tr><td>1</td><td>-2</td><td>-2</td></tr><tr><td>1</td><td>-3</td><td>-2</td></tr></table> | 1   | -1 | -2                                                    | 1 | -2 | -2             | 1 | -3 | -2 |  | <table><tr><td>-1</td></tr><tr><td>-3</td></tr><tr><td>-5</td></tr></table> | -1 | -3 | -5 | $\tanh\{3\}$ | <table><tr><td>0</td></tr><tr><td></td></tr><tr><td>-1</td></tr></table> | 0 |  | -1 | $y_1$<br>$y_2$<br>$y_3$ |
| 1                                                                                                                                          | -1  | -2 |                                                       |   |    |                |   |    |    |  |                                                                             |    |    |    |              |                                                                          |   |  |    |                         |
| 1                                                                                                                                          | -2  | -2 |                                                       |   |    |                |   |    |    |  |                                                                             |    |    |    |              |                                                                          |   |  |    |                         |
| 1                                                                                                                                          | -3  | -2 |                                                       |   |    |                |   |    |    |  |                                                                             |    |    |    |              |                                                                          |   |  |    |                         |
| -1                                                                                                                                         |     |    |                                                       |   |    |                |   |    |    |  |                                                                             |    |    |    |              |                                                                          |   |  |    |                         |
| -3                                                                                                                                         |     |    |                                                       |   |    |                |   |    |    |  |                                                                             |    |    |    |              |                                                                          |   |  |    |                         |
| -5                                                                                                                                         |     |    |                                                       |   |    |                |   |    |    |  |                                                                             |    |    |    |              |                                                                          |   |  |    |                         |
| 0                                                                                                                                          |     |    |                                                       |   |    |                |   |    |    |  |                                                                             |    |    |    |              |                                                                          |   |  |    |                         |
|                                                                                                                                            |     |    |                                                       |   |    |                |   |    |    |  |                                                                             |    |    |    |              |                                                                          |   |  |    |                         |
| -1                                                                                                                                         |     |    |                                                       |   |    |                |   |    |    |  |                                                                             |    |    |    |              |                                                                          |   |  |    |                         |
|                                                                                                                                            |     |    | $\approx$                                             |   |    |                |   |    |    |  |                                                                             |    |    |    |              |                                                                          |   |  |    |                         |

3 Level Quantization  
for hand calculation 🖋️

| Integer   | Sigmoid | Tanh |
|-----------|---------|------|
| $\geq 2$  | 1       | 1    |
| -1, 0, 1  | 0.5     | 0    |
| $\leq -2$ | 0       | -1   |

-1

# Activation (ReLU, Sigmoid, Tanh)

## Exercise 18

|    |                           |    |                                       |     |
|----|---------------------------|----|---------------------------------------|-----|
| 1  | $\tanh\{3\}$<br>$\approx$ | 0  | Sigmoid<br>$\sigma\{3\}$<br>$\approx$ | 0.5 |
| 2  |                           |    |                                       |     |
| -3 |                           | -1 |                                       | 0.5 |

3 Level Quantization  
for hand calculation 🖋️

| Integer   | Sigmoid | Tanh |
|-----------|---------|------|
| $\geq 2$  | 1       | 1    |
| -1, 0, 1  | 0.5     | 0    |
| $\leq -2$ | 0       | -1   |

0.5

# Activation (ReLU, Sigmoid, Tanh)

## Exercise 19

|    |                   |   |                                       |     |
|----|-------------------|---|---------------------------------------|-----|
| 1  | ReLU<br>$\approx$ | 1 | Sigmoid<br>$\sigma\{3\}$<br>$\approx$ | 0.5 |
| 2  |                   | 2 |                                       | 1   |
| -3 |                   |   |                                       |     |

3 Level Quantization  
for hand calculation 🖋️

| Integer   | Sigmoid | Tanh |
|-----------|---------|------|
| $\geq 2$  | 1       | 1    |
| -1, 0, 1  | 0.5     | 0    |
| $\leq -2$ | 0       | -1   |

0.5

# Activation (ReLU, Sigmoid, Tanh)

## Exercise 20

|    |               |     |
|----|---------------|-----|
| 0  | Sigmoid       | 0.5 |
| 1  | $\sigma\{5\}$ |     |
| -1 | $\approx$     | 0.3 |

5 Level Quantization  
for hand calculation 🖋️

| Integer   | Sigmoid |
|-----------|---------|
| $\geq 2$  | 1       |
| 1         | 0.7     |
| 0         | 0.5     |
| -1        | 0.3     |
| $\leq -2$ | 0       |

0.7

# Activation (ReLU, Sigmoid, Tanh)

## Exercise 21

cise 21

|     |     |    |         |               |     |       |  |
|-----|-----|----|---------|---------------|-----|-------|--|
|     |     |    | 3       | $x_1$         |     |       |  |
|     |     |    | 2       | $x_2$         |     |       |  |
| $w$ | $b$ | 1  | Sigmoid |               |     |       |  |
| 1   | 1   | -2 | 3       | $\sigma\{5\}$ | 1   | $y_1$ |  |
| 1   | 1   | -4 | 1       | $\approx$     | 0.7 | $y_2$ |  |
| 1   | 1   | -6 | -1      |               |     | $y_3$ |  |

## 5 Level Quantization for hand calculation 🖋️

| Integer   | Sigmoid |
|-----------|---------|
| $\geq 2$  | 1       |
| 1         | 0.7     |
| 0         | 0.5     |
| -1        | 0.3     |
| $\leq -2$ | 0       |

0.3

# Activation (ReLU, Sigmoid, Tanh)

## Exercise 22

|    |         |      |
|----|---------|------|
| 0  | Tanh{5} | 0    |
| 1  |         |      |
| -1 |         | -0.8 |

5 Level Quantization  
for hand calculation 🖋️

| Integer   | Sigmoid | Tanh |
|-----------|---------|------|
| $\geq 2$  | 1       | 1    |
| 1         | 0.7     | 0.8  |
| 0         | 0.5     | 0    |
| -1        | 0.3     | -0.8 |
| $\leq -2$ | 0       | -1   |

0.8

# Activation (ReLU, Sigmoid, Tanh)

## Exercise 23

cise 23

|     |     |    |    |                                 |     |       |       |
|-----|-----|----|----|---------------------------------|-----|-------|-------|
|     |     |    | 3  | $X_1$                           |     |       |       |
|     |     |    | 2  | $X_2$                           |     |       |       |
| $w$ | $b$ | 1  |    |                                 |     |       |       |
| 1   | 1   | -2 | 3  | $\text{Tanh}\{5\}$<br>$\approx$ | 1   | $Y_1$ |       |
| 1   | 1   | -4 | 1  |                                 | 0.8 | $Y_2$ |       |
| 1   | 1   | -6 | -1 |                                 |     |       | $Y_3$ |

5 Level Quantization  
for hand calculation 🖋️

| Integer   | Sigmoid | Tanh |
|-----------|---------|------|
| $\geq 2$  | 1       | 1    |
| 1         | 0.7     | 0.8  |
| 0         | 0.5     | 0    |
| -1        | 0.3     | -0.8 |
| $\leq -2$ | 0       | -1   |

-0.8

# Activation (ReLU, Sigmoid, Tanh)

## Exercise 24

|     |    |     | <div>3<br/>2</div> | $X_1$<br>$X_2$                        |     |  |  |
|-----|----|-----|--------------------|---------------------------------------|-----|--|--|
| $w$ |    | $b$ | 1                  |                                       |     |  |  |
| 1   | 0  | 0   | 3                  | Sigmoid<br>$\sigma\{5\}$<br>$\approx$ | 1   |  |  |
| 0   | 1  | 0   | 2                  |                                       | 1   |  |  |
| 1   | 1  | 0   | 5                  |                                       | 1   |  |  |
| -1  | -1 | 0   | -5                 |                                       | 0   |  |  |
| 1   | -1 | 0   | 1                  |                                       | 0.7 |  |  |
| -1  | 1  | 0   | -1                 |                                       |     |  |  |
| -1  | 0  | 0   | -3                 |                                       | 0   |  |  |
| 0   | -1 | 0   | -2                 |                                       | 0   |  |  |
| 0   | 0  | 0   | 0                  |                                       | 0.5 |  |  |

0.3



# Activation (ReLU, Sigmoid, Tanh)

## Exercise 25

|     |       |     | <table><tr><td>3</td><td><math>x_1</math></td></tr><tr><td>2</td><td><math>x_2</math></td></tr></table> | 3                               | $x_1$ | 2 | $x_2$ |  |  |  |
|-----|-------|-----|---------------------------------------------------------------------------------------------------------|---------------------------------|-------|---|-------|--|--|--|
| 3   | $x_1$ |     |                                                                                                         |                                 |       |   |       |  |  |  |
| 2   | $x_2$ |     |                                                                                                         |                                 |       |   |       |  |  |  |
| $w$ |       | $b$ | 1                                                                                                       | $\text{Tanh}\{5\}$<br>$\approx$ |       |   |       |  |  |  |
| 1   | 0     | 0   | 3                                                                                                       |                                 | 1     |   |       |  |  |  |
| 0   | 1     | 0   | 2                                                                                                       |                                 | 1     |   |       |  |  |  |
| 1   | 1     | 0   | 5                                                                                                       |                                 | 1     |   |       |  |  |  |
| -1  | -1    | 0   | -5                                                                                                      |                                 | -1    |   |       |  |  |  |
| 1   | -1    | 0   | 1                                                                                                       |                                 | 0.8   |   |       |  |  |  |
| -1  | 1     | 0   | -1                                                                                                      |                                 |       |   |       |  |  |  |
| -1  | 0     | 0   | -3                                                                                                      |                                 | -1    |   |       |  |  |  |
| 0   | -1    | 0   | -2                                                                                                      |                                 | -1    |   |       |  |  |  |
| 0   | 0     | 0   | 0                                                                                                       |                                 | 0     |   |       |  |  |  |

-0.8